

→ ↺ matlab.mathworks.com

HOME PLOTS APPS **FIGURE** FORMAT TOOLS

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Shaik

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FILE

Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

LABELS AND ANNOTATIONS

FILE

/ > MATLAB Drive

Files

Name		

Workspace

Name	Value	Size	Class
Bit_Position	[1;2;3;4;5]	5×1	double
encoded_...	[1,1,1,0,1,1,1,...]	1×10	double
encoded_...	[1,1;1,0;1,1;1,...]	5×2	double
i	5	1×1	double
Input_Bit	[1;0;0;1;1]	5×1	double
input_bits	[1,0,0,1,1]	1×5	double
Output_E...	5×1 cell	5×1	cell
output_pair	1×5 cell	1×5	cell
T	5×3 table	5×3	table
trellis	1×1 struct	1×1	struct

untitled * X

```
17 Bit_Position = (1:length(input_bits));
18 Input_Bit = input_bits;
19 Output_Encoded = output_pair;
20 T = table(Bit_Position,Input_Bit,Output_Encoded);
21 disp('Convolutional Encoding Table');
22 disp(T);
23 % Plot
24 figure;
25 subplot(2,1,1)
26 stairs(input_bits,'LineWidth',1.5)
27 ylim([-0.2 1.2])
28 title('Input Bit Sequence')
29 xlabel('Bit Position')
30 ylabel('Bit Value')
31 grid on
32 subplot(2,1,2)
33 stairs(encoded_bits,'LineWidth',1.5)
34 ylim([-0.2 1.2])
35 title('Encoded Output Bit Sequence')
36 xlabel('Bit Position')
37 ylabel('Bit Value')
38 grid on
```

Command Window

New to MATLAB? See resources for [Getting Started](#)

3	0	{ '11' }
4	1	{ '11' }
5	1	{ '01' }

>>

Figure 1 X

32°C Mostly sunny

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10:52 10-08-2026